

Total Questions: 35

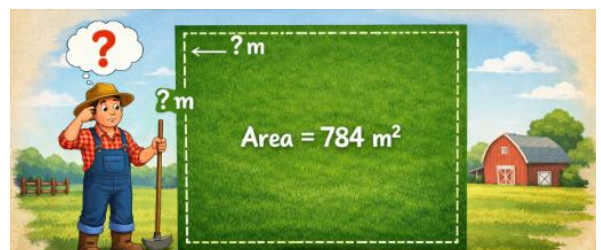
LEVEL – II

Time: 1.5 hr.
Total Marks: 70

SECTION - 1	- This section contains FIFTEEN (15) questions. Each question has FOUR options. ONLY ONE of these four options is the correct answer. - For each correct answer you will get TWO marks. There are NO NEGATIVE marks for wrong answers.
SECTION - 2	- This section contains EIGHT (08) questions. Each question has FOUR options. ONE OR MORE THAN ONE of these four option(s) is(are) correct answer(s). - For each correct answer you will get TWO marks, if all the options are correct. If the correct answers chosen are less than the actual correct answers, partial marking (basis on weightage) will be given. - There are NO NEGATIVE marks for wrong answers.
SECTION - 3	- This section contains NINE (09) questions of PASSAGE TYPE OR COMPREHENSION TYPE. - Under each paragraph there will be three questions that has FOUR options. ONLY ONE of these four options is the correct answer. - For each correct answer you will get TWO marks. There are NO NEGATIVE marks for
SECTION - 4	- This section contains THREE (03) questions. These questions are MATCHING TYPE BASED QUESTIONS. - For each correct answer you will get TWO marks. There are NO NEGATIVE marks for wrong answers

Section - 1

- Which of the following expressions represents a linear polynomial in a real-life cost model?
 (A) $3x^2 + 2$ (B) $5x + 7$
 (C) $x^3 - 1$ (D) 4
- A farmer wants to make a square field of area 784 m^2 . What will be the length of each side?



- (A) 24 m (B) 26 m
 (C) 28 m (D) 30 m

3. A group of 6 workers can complete a task in 12 days. If the number of workers increases to 12, how many days will it take to complete the same task?

(A) 3 (B) 6
(C) 12 (D) 24

4. A triangular prism has base triangle with sides 13, 14, 15 cm. Length of prism = 10 cm. Find the Volume.

(A) 840 cm^3 (B) 8400 cm^3
(C) 2100 cm^3 (D) 420 cm^3

5. A shopkeeper marks an item 25% above its cost price. During a festival, he gives a discount of 10% on the marked price. What is his overall profit percentage?

(A) 12.5% (B) 10%
(C) 8% (D) 15%

6. A microbiologist observes that a bacteria culture triples every hour. If initially there are 3^2 bacteria, what will be the population after 4 hours?

(A) 3^4 (B) 3^5
(C) 3^6 (D) 3^8

7. A tent shaped like a cone is observe(D) How many vertices does it have?



(A) 0 (B) 1
(C) 2 (D) 3

8. A car travels $\frac{5}{6}$ of a journey in 2 hours. Remaining journey is completed in $\frac{1}{3}$ hour. What fraction of journey is completed per hour in last phase?

(A) $\frac{1}{2}$ (B) $\frac{1}{6}$
(C) $\frac{1}{3}$ (D) $\frac{2}{3}$

9. A tent maker designs a conical tent for a camping site. The vertical height of the tent is 12 m and the base radius is 5 m. He needs to calculate the slant height to estimate fabric length. The slant height of the tent is:

(A) 11 m (B) 13 m
(C) 17 m (D) 15 m

10. The LCM of two numbers is 60 and their HCF is 5. If one number is 15, find the other.

(A) 10 (B) 20
(C) 30 (D) 12

11. A shopkeeper calculates profit using $(x + 2)(x + 3)$. What is the coefficient of x in the expanded form?

(A) 5 (B) 6
(C) 3 (D) 2

12. A number when multiplied by 5 and reduced by 12 gives the same result as when it is increased by 18 and then multiplied by 2. Find the number.

(A) 14 (B) 16
(C) 18 (D) 20

13. An aquarium is shaped like a rectangular prism. How many vertices does it have?



(A) 6 (B) 8
(C) 10 (D) 12

14. A machine produces 50 items in 5 hours. How many items will it produce in 8 hours?

(A) 70 (B) 75
(C) 80 (D) 85

15. A number is increased by 20% and then decreased by 20%. What is the net change?

- (A) No change (B) 4% increase
(C) 4% decrease (D) 2% decrease

Section - 2

16. Two numbers are given: $-\frac{4}{7}$ and $\frac{3}{5}$. Their sum, difference, and product are calculated. Which statements are true?

- (A) Sum is rational
(B) Product is rational
(C) Difference is irrational
(D) Closure property holds

17. Two spheres have radii in ratio 2 : 3.

Which are correct?

- (A) Surface area ratio = 4 : 9
(B) Volume ratio = 8 : 27
(C) Surface area ratio = 2 : 4
(D) Volume ratio = 4 : 9

18. A number is divisible by 9 if sum of digits is divisible by 9. Consider numbers: 729, 918, 1008, 1111. Which are divisible by 9?

- (A) 729 (B) 918
(C) 1008 (D) 1111

19. A ratio is written as $(\frac{5}{2})^{-3}$. Which are correct?

- (A) It becomes $(\frac{2}{5})^3$
(B) It equals $\frac{8}{125}$
(C) Negative exponent inverts fraction
(D) It equals $125/8$

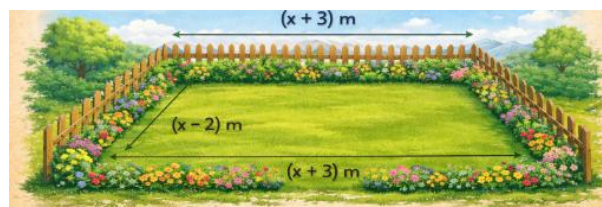
20. A dealer marks his goods 40% above cost price and gives a discount of 20%. Which are correct?

- (A) He makes profit
(B) Profit % = 12%

(C) Profit % = 20%

(D) Selling price > Cost price

21. A rectangular garden has length $(x + 3)$ m and breadth $(x - 2)$ m. If perimeter is 30 m, which are correct?



(A) $x = 7$ (B) Length = 10 m

© Breadth = 5 m (D) Area = 50 m²

22. Which of the following numbers can NOT be perfect squares based on unit digit?

- (A) 132 (B) 145
(C) 278 (D) 361

23. In a real-life pricing model, $5x \cdot 2y^2$ is used. Which are correct?

- (A) Result is $10xy^2$
(B) It is a monomial
(C) Degree is 3
(D) Coefficients are added

Section - 3

Passage 1:

A city park has a circular lawn of radius 14 m. Around it, a walking path of uniform width 3 m is constructed forming a ring (annulus). A decorative sector of angle 90° is marked for a fountain area inside the lawn.

24. What is the area of the walking path?

- (A) 264π (B) 93π
(C) 30π (D) 324π

25. What is the area of the fountain sector?

- (A) 49π (B) 98π
(C) 154π (D) 196π

26. If the cost of grassing is ₹5 per m^2 , what is the cost of covering the remaining lawn (excluding fountain)?

- (A) ₹3080 (B) ₹2310
(C) ₹3850 (D) ₹4200

PASSAGE-2

The sum of ages of a father and son is 70 years. Five years ago, the father's age was 4 times the son's age.

Let present age of son = x .

27. Which equation represents the condition 5 years ago?

- (A) $(70 - x - 5) = 4(x - 5)$
(B) $(70 - x) = 4x$
(C) $(70 - x + 5) = 4(x + 5)$
(D) $(70 - x - 5) = x - 5$

28. Present age of son is:

- (A) 10 years (B) 17 years
(C) 20 years (D) 12 years

29. Present age of father is:

- (A) 60 years
(B) 53 years
(C) 50 years
(D) 65 years

PASSAGE-3

A construction engineer needs to design a square platform of area 1296 m^2 and calculates its side using the division method of square roots.

Later, he measures a smaller decorative square area of 17.64 m^2 using decimal square root techniques.

To verify accuracy, he estimates square roots of nearby numbers like 256 and 3844. Finally, he ensures all values satisfy properties of square roots before implementation.

30. Using the division method, the side of the platform is:

- (A) 34 m (B) 35 m
(C) 36 m (D) 38 m

31. Using estimation, $\sqrt{256}$ equals:

- (A) 14 (B) 15
(C) 16 (D) 18

32. Which of the following is correct for $\sqrt{3844}$?

- (A) 60 (B) 62
(C) 64 (D) 66

Section - 4

33. A data analyst observes cube patterns in numbers. Match the following.

Column I (Situation)	Column II (Value)
(a) Cube of 15 in large data modeling	(i) 3375
(b) A number whose cube is 64	(ii) 4
(c) Cube of -7 in negative growth pattern	(iii) -343
(d) Volume of cube with side 13 cm	(iv) 2197

- (A) a-ii, b-i, c-iv, d-iii
(B) a-ii, b-iv, c-iii, d-i
(C) a-i, b-ii, c-iii, d-iv
(D) a-iii, b-i, c-iv, d-ii

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Column I (Equations)	Column II (Final Values of x)
(a) $\frac{x}{2} - \frac{1}{5} = \frac{x}{3} + \frac{1}{4} + 1$	(i) $x = -8$
(b) $\frac{x}{3} + 2 = \frac{x}{6} + 5$	(ii) $x = 8.7$
(c) $\frac{x}{4} - 3 = \frac{x}{2} - 1$	(iii) $x = 1/2$
(d) $\frac{(x+1)}{(2x+3)} = \frac{3}{8}$	(iv) $x = 18$

- (A) a-ii, b-i, c-iii, d-iv
 (B) a-ii, b-iv, c-i, d-iii
 (C) a-i, b-iv, c-iii, d-ii
 (D) a-ii, b-i, c-iv, d-iii

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Column I (Situations)	Column II (Values)
(a) If 8 workers complete a work in 12 days, find the number of days required for 16 workers.	(i) 4
(b) If a car takes 6 hours to reach a place at a certain speed, find the time taken if the speed is doubled	(ii) 6
(c) If 12 men can complete a task in 10 days, find the days required for 6 men.	(iii) 20
(d) If 4 taps fill a tank in 8 hours, find the time taken by 8 taps.	(iv) 3

- (A) a-iv, b-i, c-ii, d-iii
 (B) a-ii, b-iv, c-iii, d-i
 (C) a-i, b-ii, c-iii, d-iv
 (D) a-ii, b-i, c-iv, d-iii